The fixed-kernel regime of a margin-conditioned Bayesian neural-network prior

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Abstract

We study the width limit of a fully connected tanh network on the Boolean cube with an appended constant input, under the Gaussian scaling $\mathrm{Var}(W_{ij}^1) = N^{1-\gamma}/d$, $\mathrm{Var}(W_{ij}^s) = N^{-\gamma}$ for later hidden layers, and $\mathrm{Var}(W_i^L) = N^{-\gamma-1}$ for the readout. The Bayesian prior is conditioned on an output margin at every Boolean input; training then conditions on observed signs. For every fixed $\gamma < 1$, dimension, and depth, hidden-feature covariances concentrate at speed N around an explicit iterated-arcsine kernel. For $0 < \gamma < 1$, outputs satisfy a Gaussian large-deviation principle at speed N^γ, and the actual posterior concentrates on compatible Boolean functions of minimum kernel margin energy. At $\gamma = 0$ the limit is a Gaussian-process classifier with the global margin restriction; at $\gamma < 0$ the margin disappears and the limit is a Gaussian sign classifier. These establish a fixed-kernel regime under the original prediction conventions, while distinguishing it from Gaussian-process regression. We derive exact parity prior rates, a strictly paired kernel spectrum, the complete set of unconditioned minimum-energy sign functions, and classification lower bounds. A prediction counts as successful only when the posterior probability of its correct label exceeds 2/3. For $\gamma \le 0$ and depth $O(d)$, we prove that the failure fraction is exactly the unseen fraction in the width limit, giving exponential sample complexity when the target is parity on d bits. The stronger claim that every fixed $\gamma < 1$ requires exponentially many examples for a k-bit parity is not established: a small parity eigenvalue alone does not imply that claim for this posterior.

1 Model, posterior, and order of limits

Fix $d \ge 1$, depth $L \ge 2$, margin $a > 0$, and a real exponent $\gamma < 1$. Inputs are $x \in \mathcal{X} = \{-1, 1\}^d$, with $h^0(x) = (1, x)$. All hidden layers have width N and

$$h^s(x) = \tanh(W^s h^{s-1}(x)), \quad 1 \le s \le L-1, \qquad z_N(x) = W^L h^{L-1}(x).$$

The activation acts coordinatewise. There are no biases after the appended constant input. All weights are independent and centered Gaussian, with variances

$$\mathrm{Var}(W_{ij}^1) = \frac{N^{1-\gamma}}{d}, \qquad \mathrm{Var}(W_{ij}^s) = N^{-\gamma} \ (1 < s < L), \qquad \mathrm{Var}(W_i^L) = N^{-\gamma-1}. \tag{1}$$

This is the strong-prior scaling line in the source notes [1], continued below $\gamma = 1$. The main statements are specific to this line. Changing the relative layer variances or inserting additional biases can change the limiting kernel.

Let $Z_N = (z_N(x))_{x \in \mathcal{X}}$. The prior on Boolean functions is obtained by first conditioning on

$$A_a = \{|Z_N(x)| > a \text{ for all } x \in \mathcal{X}\}$$

and then taking $g(x) = \operatorname{sign} Z_N(x)$. A training sample $S = ((X_i, f(X_i)))_{i=1}^n$ supplies noiseless signs. Write $\mathcal{G}_S = \{g : g(X_i) = f(X_i) \text{ for all } i\}$. For $A_g = \{g(x)Z_N(x) > a \text{ for every } x\}$, the exact posterior is

$$\pi_N(g \mid S) = \frac{\Pr(A_g)}{\sum_{h \in \mathcal{G}_S} \Pr(A_h)} \quad (g \in \mathcal{G}_S), \tag{2}$$

with zero mass outside $\mathcal{G}_S$. The normalization introduced by the initial global-margin conditioning cancels in this formula.

At an input x, let $p_N(x \mid S, f) = \pi_N\{g : g(x) = f(x) \mid S\}$. We count a prediction as successful only when $p_N(x \mid S, f) > 2/3$. Our primary error criterion is therefore the fraction of inputs failing this confidence requirement:

$$F_N(S, f) = \frac{1}{2^d} \sum_{x \in \mathcal{X}} \mathbf{1}\{p_N(x \mid S, f) \le 2/3\}. \tag{3}$$

As a useful auxiliary quantity, define the average posterior probability of a wrong label:

$$R_N(S, f) = \frac{1}{2^d} \sum_{x \in \mathcal{X}} \pi_N\{g : g(x) \ne f(x) \mid S\}. \tag{4}$$

This is also the expected test error of a Boolean function drawn from the posterior. Pointwise comparison of the wrong-label probability on the failing and successful inputs gives

$$\max\{0, (3R_N - 1)/2\} \le F_N \le \min\{1, 3R_N\}. \tag{5}$$

Indeed, failing inputs each contribute at least $1/3$ to R_N, while successful inputs contribute less than $1/3$ and all inputs contribute at most one. The same linear bounds hold after averaging over samples. In particular, posterior error near $1/2$ forces confidence failure on at least about a quarter of inputs.

The majority rule chooses the more probable sign at each input and is a different performance criterion. Its error is at most twice (4); the reverse implication fails. A posterior that assigns probability 0.5001 to every correct sign can have a perfect majority classifier and posterior error close to $1/2$.

Every width limit in this paper takes $N \to \infty$ with d, L, γ, a fixed. Sample-complexity statements then study the resulting family as d, k, n vary, where k is the parity size. We do not exchange these limits or claim quantitative uniformity in d or L.

2 The regime $\gamma < 1$: a frozen kernel and three posterior limits

Throughout this section the activation is tanh, the depth $L \ge 2$ and input dimension $d \ge 1$ are fixed, and the width tends to infinity. Write $\mathcal{X} = \{-1, 1\}^d$, $M = 2^d$, and let $a > 0$ be the required margin. All limits below hold at each fixed $\gamma < 1$; uniformity in dimension, depth, or $\gamma \uparrow 1$ is not asserted. Define

$$K^{(0)}(x, y) = \frac{1 + x \cdot y}{d + 1}, \qquad K^{(s+1)}(x, y) = \frac{2}{\pi} \arcsin K^{(s)}(x, y), \qquad K = K^{(L-1)}. \tag{6}$$

Thus the limiting kernel uses the saturated activation. It is not the unit-variance, finite-gain tanh kernel.

Theorem 2.1 (Exponential freezing of the feature kernel)**.** *Let $h_i^s(x)$ denote the activation of hidden neuron i in layer s, and set*

$$Q_N^s(x,y) = \frac{1}{N}\sum_{i=1}^{N} h_i^s(x)h_i^s(y).$$

For every $\epsilon > 0$ and each $1 \le s \le L-1$, there are constants $c, C > 0$ and N_0 such that

$$\Pr\left\{\max_{x,y} |Q_N^s(x,y) - K^{(s)}(x,y)| > \epsilon\right\} \le Ce^{-cN} \qquad (N \ge N_0).$$

The matrix K is positive definite. Conditional on all hidden layers, the output vector $Z_N = (z_N(x))_{x\in\mathcal{X}}$ has the exact law

$$Z_N \mid (\textit{hidden layers}) \sim \mathcal{N}(0, N^{-\gamma}Q_N^{L-1}). \tag{7}$$

Consequently $N^{\gamma/2}Z_N$ converges in distribution to $G \sim \mathcal{N}(0,K)$.

Proof. Put $b_N = N^{(1-\gamma)/2} \to \infty$. Conditional on the preceding layer, different preactivation rows are independent Gaussian vectors with covariance $b_N^2 Q_N^{s-1}$. For the first layer, replace this covariance matrix by the deterministic matrix $b_N^2 R_0$, where $R_0(x,y) = (1 + x\cdot y)/d$; its correlation matrix is $K^{(0)}$.

If $U \sim \mathcal{N}(0,q)$, then bounding the Gaussian density by its maximum and integrating $1 - \tanh v$ gives

$$\mathbb{E}|\tanh(bU) - \operatorname{sign} U| \le \frac{2\log 2}{b\sqrt{2\pi q}}.$$

For a centered Gaussian pair with correlation r, the expected product of its signs is $2\arcsin(r)/\pi$. Thus, uniformly when coordinate variances are bounded below, the expected product of the two saturated activations differs from this arcsine expression by $O(b_N^{-1})$. The appended constant input implies $|K^{(s)}(x,y)| < 1$ for distinct x, y. In a sufficiently small fixed neighborhood of $K^{(s)}$, the map sending a covariance matrix to its normalized arcsine matrix is Lipschitz, and all diagonal variances are bounded below. Finally, conditional Hoeffding bounds for the N independent products in $[-1,1]$ give

$$\Pr\left\{\max_{x,y} |Q_N^s(x,y) - \mathbb{E}[Q_N^s(x,y) \mid \text{preceding layer}]| > t \,\middle|\, \text{preceding layer}\right\} \le 2M^2 e^{-Nt^2/2}.$$

Induction over the fixed number of layers proves the concentration statement, absorbing the deterministic $O(b_N^{-1})$ error by taking N sufficiently large.

For positive definiteness, $K^{(1)}$ is the Gram matrix of the functions $u \mapsto \operatorname{sign}(u\cdot(1,x))$ under a nondegenerate Gaussian measure. They are linearly independent: in a vanishing linear combination, cross the hyperplane $u\cdot(1,x) = 0$ at a point on none of the other finitely many hyperplanes. Only the corresponding term changes sign, so its coefficient is zero. Such a point exists because the augmented vectors $(1,x)$ are pairwise nonparallel. The positive power-series coefficients of the arcsine function and the Schur product theorem then imply $K^{(s+1)} \succeq (2/\pi)K^{(s)}$, preserving strict positive definiteness. The independent Gaussian readout proves (7), and convergence follows. □

2.1 Positive γ: the actual posterior is a kernel-energy learner

For a Boolean function g define

$$E_K(g) = \min\left\{\frac{1}{2}z^T K^{-1} z : g(x)z(x) \ge a \text{ for every } x \in \mathcal{X}\right\}. \tag{8}$$

This minimum is finite and has a unique minimizing vector, by strict convexity and coercivity.

Theorem 2.2 (Kernel posterior limit)**.** *For each fixed $0 < \gamma < 1$, Z_N satisfies a large-deviation principle with speed N^γ and good rate function $I(z) = z^T K^{-1} z/2$. In particular,*

$$\lim_{N\to\infty} -N^{-\gamma} \log \Pr\{g(x)Z_N(x) > a \text{ for every } x\} = E_K(g). \tag{9}$$

Let $\mathcal{G}_S$ be the Boolean functions consistent with the observed training labels and let $E_S = \min_{h\in\mathcal{G}_S} E_K(h)$. Under the all-input-margin prior followed by noiseless sign conditioning,

$$\lim_{N\to\infty} -N^{-\gamma} \log \Pr\{\operatorname{sign} Z_N = g \mid \text{all margins and } S\} = E_K(g) - E_S \qquad (g \in \mathcal{G}_S). \tag{10}$$

The posterior therefore concentrates on the functions in $\operatorname{argmin}_{g\in\mathcal{G}_S} E_K(g)$.

Proof. On a common probability space let

$$Z_N = N^{-\gamma/2}(Q_N^{L-1})^{1/2}\xi, \qquad Y_N = N^{-\gamma/2}K^{1/2}\xi, \qquad \xi \sim \mathcal{N}(0, I_M),$$

with ξ independent of the hidden layers. The square-root map is locally Lipschitz near the positive definite K. For fixed $\delta, \epsilon > 0$, Theorem 2.1 consequently gives

$$\Pr\{\|Z_N - Y_N\| > \delta\} \le C_\epsilon e^{-c_\epsilon N} + \Pr\left\{\|\xi\| > \frac{\delta N^{\gamma/2}}{C\epsilon}\right\}.$$

Divide logarithms by N^γ, first let $N \to \infty$, and then let $\epsilon \downarrow 0$. Because $\gamma < 1$, the resulting upper limit is $-\infty$. Thus Z_N is exponentially equivalent, at speed N^γ, to the elementary small-noise Gaussian family Y_N. The latter has rate I, proving the large-deviation statement.

The strict margin orthant is open and its closure is the feasible set in (8). Scaling its unique minimizer by $1 + \eta$ puts it in the open orthant and approaches the same energy as $\eta \downarrow 0$. The open lower bound and closed upper bound therefore match. Summing over the finitely many compatible sign patterns and subtracting logarithmic asymptotics proves (10). □

If the minimizing Boolean function is unique, the limiting posterior is its point mass. If there is a tie, the exponential rates do not determine the mixture weights. The result does not prescribe a uniform mixture of minimizers.

The limit is governed by a fixed kernel, but its optimization problem retains the constraint $|z(x)| \ge a$ at every unseen input. Its feasible set is

$$\{z : f(x_i)z(x_i) \ge a \text{ at observed inputs}, \quad |z(x)| \ge a \text{ at every input}\}.$$

This is a nonconvex union of orthants. Dropping the unseen-input constraints produces a different learner. In particular, ordinary kernel interpolation, Gaussian-process regression, and the usual hard-margin support-vector machine do not describe this posterior.

2.2 Zero and negative γ

Proposition 2.3 (The remaining fixed-kernel limits)**.** *At $\gamma = 0$, the posterior on Boolean functions converges to that of $G \sim \mathcal{N}(0, K)$ conditioned on $|G(x)| > a$ at every input and on the observed training signs. At $\gamma < 0$, it converges to the sign law of the same Gaussian vector conditioned only on observed training signs.*

Proof. The normalized outputs converge weakly to the nondegenerate G. At $\gamma = 0$, every relevant conditioning event has boundary in a finite union of hyperplanes of Gaussian probability zero, and its probability is strictly positive. Conditional probabilities converge. At $\gamma < 0$, the normalized margin is $aN^{\gamma/2} \to 0$. The probability that any normalized coordinate falls within this margin tends to zero. Every prescribed finite set of observed signs has positive limiting probability, so the same conditional convergence argument applies. □

2.3 What the kernel limit says about parity

Cube symmetries diagonalize K in the parity basis. If $\chi_U(x) = \prod_{i \in U} x_i$ and λ_U is its eigenvalue for the normalized operator $M^{-1}K$, then

$$E_K(\chi_U) = \frac{a^2}{2\lambda_U}. \tag{11}$$

Indeed $a\chi_U$ is feasible, and its gradient $K^{-1}(a\chi_U)$ is a positive multiple of χ_U. Its inner product with every feasible displacement is nonnegative; the first-order criterion for convex minimization proves optimality.

Equation (11) is an exact prior-rate calculation. It is not a sample-complexity lower bound. For $0 < \gamma < 1$, that question requires comparison with the energies of all compatible Boolean functions; for $\gamma = 0$ it concerns globally margin-conditioned Gaussian orthant probabilities; for $\gamma < 0$ it concerns zero-margin Gaussian sign probabilities. A small parity eigenvalue by itself does not answer any of these posterior-learning questions. The regression subspace argument also does not apply: the full-cube conditioning need not leave the posterior mean in the span of training kernel sections. Moreover a lower bound stated in the parity size k must specify the ambient dimension, depth, input distribution, order of limits, and whether error means posterior misclassification probability or majority-vote error.

3 Classification bounds and the limits of eigenvalue arguments

3.1 The spectrum and the first functions eliminated by data

Proposition 3.1 (Paired spectrum and minimum-energy signs)**.** *Let λ_r be the eigenvalue of $M^{-1}K$ on degree-r parities. For the kernel (6),*

$$\lambda_0 = \lambda_1 > \lambda_2 = \lambda_3 > \lambda_4 = \lambda_5 > \cdots > 0,$$

with a final unpaired λ_d when d is even. The minimum of $E_K(g)$ over all Boolean functions is $a^2/(2\lambda_0)$, attained exactly by the constants and signed coordinate functions $g \in \{1, -1, x_1, -x_1, \ldots, x_d, -x_d\}$. For every finite width, all these functions have identical unconditioned full-margin prior probabilities.

Proof. Put $D = d + 1$ and write F for the $(L-1)$-fold composition of $t \mapsto 2\arcsin(t)/\pi$. Its power series is $F(t) = \sum_{m \text{ odd}} c_m t^m$, with every $c_m > 0$ and $\sum_m c_m = 1$. Positivity follows from the arcsine series and is preserved under composition. First consider the kernel $F(u \cdot v/D)$ on the full D-dimensional cube. The degree-r Fourier eigenvalue of $(u \cdot v/D)^m$ is the probability $p_m(r)$ that a simple random walk, starting at one cube vertex and flipping a uniformly chosen coordinate at each step, reaches a specified vertex at Hamming distance r after m steps. This follows by expanding the power: a term survives Fourier averaging exactly when the coordinates chosen an odd number of times form the specified support.

These probabilities decrease with distance within each parity class. Indeed, for $0 \le r \le D - 2$,

$$\begin{aligned} D\big[p_{m+1}(r) - p_{m+1}(r+2)\big] &= r\big[p_m(r-1) - p_m(r+1)\big] \\ &\quad + (D - r - 2)\big[p_m(r+1) - p_m(r+3)\big], \end{aligned}$$

where terms with zero coefficients at the boundaries are omitted. Induction starts from the point mass at time zero. Since $p_r(r) > 0 = p_r(r+2)$ and $c_r > 0$ for odd r, the positive odd-degree

eigenvalues of the full-cube kernel decrease strictly; its even-degree eigenvalues vanish. Restricting the distinguished coordinate to $+1$ makes the reduced degree-r eigenvalue the sum of the full-cube degree-r and degree-$(r+1)$ eigenvalues. This gives the paired ordering stated.

The largest matrix eigenvalue of K is $M\lambda_0$. Every full-margin output satisfies

$$\frac{1}{2}z^T K^{-1} z \geq \frac{\|z\|_2^2}{2M\lambda_0} \geq \frac{a^2}{2\lambda_0}.$$

Equality requires both $|z(x)| = a$ everywhere and z/a in the span of $1, x_1, \ldots, x_d$. An affine function $c + \sum_j b_j x_j$ is Boolean only if it is a constant or signed coordinate: expanding its square shows $b_i b_j = 0$ for $i \neq j$ and $c b_i = 0$, so at most one coefficient is nonzero, with magnitude one. These functions attain equality, completing the characterization.

To prove the finite-width equality of prior weights, exchange columns 0 and j of the first-layer weight matrix. Its Gaussian law is unchanged. Let P_j denote this exchange and let T_j be the cube bijection defined by $(T_j x)_j = x_j$ and $(T_j x)_i = x_j x_i$ for $i \neq j$. Then $P_j(1, x) = x_j(1, T_j x)$. The network as a function of its augmented input is odd, since all activations are odd and there are no later biases. Thus the transformed output is $x_j z_N(T_j x)$. The all-positive margin orthant is mapped bijectively to that of x_j, proving equality. Flipping the readout weights proves equality with the negative functions as well. □

Corollary 3.2 (A posterior-error lower bound from surviving coordinates)**.** *Fix $0 < \gamma < 1$ and let $f = \chi_U$ be a parity of size $k \geq 2$. If any constant or signed coordinate agrees with a training sample S, the posterior converges to the uniform distribution over all such compatible functions. In particular,*

$$\lim_{N\to\infty} R_N(S, f) = \frac{1}{2}.$$

For $d > k$ and $n \geq 1$ independent uniform training examples,

$$\liminf_{N\to\infty} \mathbb{E}_S R_N(S, f) \geq \frac{1}{2}\left[1 - \left(1 - 2^{1-n}\right)^{d-k}\right]. \tag{12}$$

Proof. If such a function survives, the compatible absolute minimizers in Proposition 3.1 exhaust the compatible minimum-energy functions. Theorem 2.2 forces the posterior onto these functions, and their exact equal prior weights make the conditional distribution uniform. Each has uniform error exactly $1/2$ against a parity of degree at least two, independently of the mixture weights. This proves the first statement.

Conditional on all samples in coordinates belonging to U, the target labels are fixed. Each of the $d - k$ irrelevant coordinates has an independent uniform sample column in $\{-1, 1\}^n$. One of its two signs agrees with all labels with probability 2^{1-n}. These events are conditionally independent. The probability that at least one such coordinate survives is consequently $1 - (1 - 2^{1-n})^{d-k}$. On this event the limiting error is $1/2$; on its complement error is nonnegative. Fatou's lemma proves the bound. □

Lemma 3.3 (Parity correlation with a halfspace)**.** *If h is a Boolean halfspace on $\{-1, 1\}^d$ and $|U| = k \geq 1$, then $|\mathbb{E}[\chi_U h]| \leq 1/\sqrt{k}$. Either inclusion or exclusion of the threshold boundary is allowed.*

Proof. Condition on coordinates outside U and flip selected input coordinates so that the halfspace is monotone in the remaining k coordinates. These flips change the parity only by a constant sign.

For a monotone Boolean function h, its discrete derivative $D_i h = (h(x_i = 1) - h(x_i = -1))/2$ is nonnegative. Thus, for every $i \in U$,

$$|\widehat{h}(U)| = |\mathbb{E}[\chi_{U\setminus\{i\}} D_i h]| \le \mathbb{E} D_i h = \widehat{h}(\{i\}).$$

Summing over the k coordinates and using Cauchy–Schwarz and Parseval gives $k|\widehat{h}(U)| \le \sqrt{k}$. Average the conditional estimate over the other coordinates. □

Corollary 3.4 (Failure to attain posterior confidence 2/3)**.** *Under the hypotheses of Corollary 3.2, if any constant or signed coordinate is consistent with S, then*

$$\liminf_{N\to\infty} F_N(S, f) \ge b_k := \max\left\{\frac{1}{4}, \frac{1}{2} - \frac{1}{2\sqrt{k}}\right\}.$$

If there are m such compatible functions, the additional bound $\liminf_N F_N(S, f) \ge 1 - 9/m$ holds. Consequently, when $d > k$ and $n \ge 1$,

$$\liminf_{N\to\infty} \mathbb{E}_S F_N(S, f) \ge b_k \left[1 - \left(1 - 2^{1-n}\right)^{d-k}\right].$$

Proof. The bound 1/4 follows from (5) and Corollary 3.2. To obtain the stronger bound for large k, write $\epsilon_N \to 0$ for the posterior mass outside the consistent absolute minimizers. The mean Boolean prediction conditioned on those minimizers is an affine function $\widetilde{m}_N(x) = c_N + \sum_j b_{N,j} x_j$. The full posterior mean $m_N(x)$ satisfies $\|m_N - \widetilde{m}_N\|_\infty \le 2\epsilon_N$. Confidence success is the event $f(x)m_N(x) > 1/3$, which implies $f(x)\widetilde{m}_N(x) > t_N := 1/3 - 2\epsilon_N > 0$ for large N.

For any affine m and any $t > 0$, set $w(x) = \mathbf{1}\{m(x) > t\} - \mathbf{1}\{m(x) < -t\}$. This is the average of two Boolean halfspaces, using the appropriate boundary conventions. By Lemma 3.3, $|\mathbb{E} fw| \le 1/\sqrt{k}$. Moreover

$$\Pr\{fm > t\} = \frac{1}{2}\mathbb{E}(|w| + fw) \le \frac{1}{2} + \frac{1}{2\sqrt{k}}.$$

Apply this to $m = \widetilde{m}_N$ and $t = t_N$. This proves the pointwise-in-sample bound without assuming convergence of the weights among tied minimizers. The expected bound follows using the event probability calculated in Corollary 3.2 and Fatou's lemma.

For the additional bound, equal weights make $\widetilde{m}_N = m^{-1}\sum_{j=1}^m g_j$ independent of N. Distinct signed constants and coordinates are orthogonal unless they are negatives of one another. Hence $\mathbb{E}\widetilde{m}_N^2 \le 1/m$. Confidence success implies $|\widetilde{m}_N| > t_N$, so Markov's inequality bounds its fraction by $1/(mt_N^2) \to 9/m$. □

More explicitly, put $D = d - k$ and $p = 2^{1-n}$ for $n \ge 1$. The number of surviving irrelevant coordinates is $\mathrm{Bin}(D, p)$. For any integer $t > 9$, the same proof gives

$$\liminf_N \mathbb{E}_S F_N(S, f) \ge (1 - 9/t)\Pr\{\mathrm{Bin}(D, p) \ge t\}.$$

Thus confidence failure can be close to one while many spurious coordinates remain compatible with the sample.

These results establish a necessary sample scale of order $\log(d - k)$ for posterior confidence in this regime. They are not exponential-in-k bounds, nor do they assert the same error for posterior majority. The spectrum also gives $E_K(\chi_U) \ge (a^2/2)\binom{d}{k}$, since $\sum_{r=0}^d \binom{d}{r}\lambda_r = 1$. Even an exponentially large parity energy when $d \ge 2k$ is a prior statement rather than a sample lower bound.

3.2 An information bound that does apply to the posterior

Proposition 3.5 (Parity-identification bound)**.** *Let $1 \le k \le d$ and $B = \binom{d}{k}$. For n independent uniform training inputs and any fixed degree-k parity f, the posterior learner in this paper satisfies*

$$\mathbb{E}_S R_N(S, f) \ge \frac{1}{2}\left(1 - \sqrt{\frac{2^n}{B}}\right) \tag{13}$$

whenever its posterior is defined. The same statement holds for its width limits and for a majority classifier whose tie rule respects coordinate permutations. Thus expected error at most $\epsilon < 1/2$ requires

$$n \ge \log_2 \binom{d}{k} + 2\log_2(1 - 2\epsilon).$$

Proof. First average the target uniformly over the B degree-k parities and fix the training inputs. The targets split into at most 2^n cells C_y according to their label vectors. Let $m_y(x) \in [-1, 1]$ denote the learner's mean signed prediction given the label vector y. For the posterior learner it is the posterior marginal mean of $g(x)$; for majority prediction it is the selected sign. Use the uniform inner product on the Boolean cube. Since $\|m_y\|_2 \le 1$, orthogonality of the parities and Bessel's inequality give

$$\sum_{U \in C_y} \langle m_y, \chi_U \rangle \le \sqrt{|C_y|}\left(\sum_{U \in C_y} \langle m_y, \chi_U \rangle^2\right)^{1/2} \le \sqrt{|C_y|}.$$

Summing over cells and applying Cauchy–Schwarz bounds average target correlation by $\sqrt{2^n/B}$. Classification error is half of one minus this correlation. Finally, coordinate permutations act transitively on degree-k supports and preserve the network prior, the all-input margin event, the sampling distribution, and the posterior rule. The expected error is therefore identical for every degree-k parity. This proves the fixed-target result as well as its stated extensions. □

This lower bound is of order $k\log(d/k)$ when $k \ll d$, and is vacuous when $k = d$. It is not exponential in k. It also illustrates why the ambient dimension matters in any statement about "an XOR of size k." For the primary confidence criterion, the same argument and (5) imply

$$\mathbb{E}_S F_N(S, f) \ge \max\left\{0, \frac{1}{4} - \frac{3}{4}\sqrt{\frac{2^n}{\binom{d}{k}}}\right\}.$$

3.3 Exponential confidence sample complexity at sufficient depth

We can prove exponential sample complexity for the actual scaling family when $\gamma \le 0$ and depth is allowed to grow linearly with dimension. This uses a uniform comparison after conditioning on all margins and any training signs. Its uniformity is essential: the conditioning event can itself be exponentially unlikely.

Lemma 3.6 (Gaussian comparison after margins and signs)**.** *For each $a \ge 0$ there is $c_a > 0$ such that, for an $M \times M$ covariance matrix K and $0 < \eta < 1/4$, the condition*

$$\|K - I\|_{\mathrm{op}} \le c_a \eta / M$$

implies the following. If B prescribes $|z_j| > a$ at every coordinate and arbitrary signs on any subset, then

$$\mathrm{TV}(\mathcal{N}(0, K)(\cdot \mid B), \mathcal{N}(0, I)(\cdot \mid B)) \le \eta.$$

When $a = 0$, the all-coordinate absolute-value condition is vacuous.

Proof. Let $Q = \mathcal{N}(0, I)(\cdot \mid B)$, $H = K^{-1} - I$, and $h = \|H\|_{\mathrm{op}}$. Conditional densities have ratio $w/\mathbb{E}_Q w$, where $w(z) = \exp(-z^T H z/2)$. Under Q, the squared coordinates are independent with the law of G^2 conditional on $|G| > a$, independently of the prescribed signs. Put $\mu_a = \mathbb{E}[G^2 \mid |G| > a]$. For a finite D_a,

$$\log \mathbb{E}[e^{tG^2} \mid |G| > a] \le D_a t \quad (0 \le t \le 1/4),$$

because the moment-generating function and its derivative are finite on this interval. With $T = \sum_j z_j^2$ and $h \le 1/2$,

$$\mathbb{E}_Q|w - 1| \le e^{D_a Mh/2} - 1, \qquad \mathbb{E}_Q w \ge e^{-\mu_a Mh/2}.$$

The second inequality uses Jensen's inequality. Thus

$$\mathrm{TV} \le \frac{\mathbb{E}_Q|w-1|}{\mathbb{E}_Q w} \le e^{\mu_a Mh/2}(e^{D_a Mh/2} - 1).$$

For $Mh \le 1$ the last expression is at most a constant depending only on a times Mh. The inverse identity gives $h \le 2\|K - I\|_{\mathrm{op}}$ when the latter norm is at most $1/2$. Choose c_a small enough to obtain the claim. □

Lemma 3.7 (Decay of saturated-kernel correlations)**.** *There is an absolute constant C such that the kernel in (6) obeys $K(x, x) = 1$ and $|K(x, y)| \le \rho$ for every distinct x, y, whenever*

$$L - 1 \ge C(\log(d + 1) + \log(1/\rho) + 1).$$

Proof. Write $T(r) = 2\arcsin(r)/\pi$. This map is odd and satisfies $0 < T(r) < r$ for $0 < r < 1$. Since $T'(r) \to \infty$ as $r \uparrow 1$, there is $r_* < 1$ for which $1 - T(r) \ge 2(1 - r)$ whenever $r_* \le r < 1$. Initially the absolute correlation of any distinct augmented cube points is at most $(d - 1)/(d + 1)$. Its distance from one therefore doubles until its iterates fall below r_*, requiring at most $O(\log(d+1)+1)$ steps. The positive power series shows $T(r)/r \le T(r_*)/r_* =: q < 1$ for $0 < r \le r_*$. A further $O(\log(1/\rho) + 1)$ steps suffice. Oddness handles negative correlations, and $T(1) = 1$ preserves the diagonal. □

Theorem 3.8 (Exact confidence failure in a deep subfamily)**.** *Fix $\gamma \le 0$ and $a > 0$. There is a constant C_a such that, if $L \ge C_a(d+1)$, then for every target and every fixed training sample S, the width-limit confidence failure is exactly*

$$\lim_{N\to\infty} F_N(S, f) = \frac{U_S}{2^d},$$

where U_S is the number of cube inputs absent from the sample. Consequently, for n independent uniform training inputs,

$$\lim_{N\to\infty} \mathbb{E}_S F_N(S, f) = (1 - 2^{-d})^n. \tag{14}$$

For $0 < \epsilon \le 1/2$, expected confidence failure at most ϵ therefore requires and is achieved by $\Theta(2^d \log(1/\epsilon))$ examples. In particular, for $d = k$ this is exponential in the size of a parity circuit.

Proof. Choose a fixed $\eta < 1/6$, for example $\eta = 1/12$. Apply Lemma 3.7 with $\rho = c\eta/M^2$, where $M = 2^d$ and c is the smaller of the comparison constants for margins a and zero. Its depth requirement is $O(d+1)$, and $\|K - I\|_{\mathrm{op}} \le (M - 1)\rho \le c\eta/M$. At $\gamma = 0$ use Lemma 3.6 with margin a; at $\gamma < 0$ use it with margin zero, in accordance with Proposition 2.3.

Under the independent Gaussian comparator, every unobserved sign is fair after conditioning. The actual limiting probability of either sign at each such input is consequently at most $1/2+\eta < 2/3$.

Every observed input has correct-label probability one. Conditional convergence and the strict separation from 2/3 then give the first assertion. Averaging over the finite set of sample sequences is legitimate; each input is unobserved with probability $(1 - 1/M)^n$. Solving the resulting equality proves the sample bound. □

This theorem lets d and the chosen depth vary after the width limit. It does not prove an exponential bound for depth fixed independently of d, nor for $0 < \gamma < 1$, where the effective Gaussian margin diverges and a fixed near-independence estimate does not control the zero-temperature posterior.

The same confidence conclusion sharpens the separate NNGP note [2]. That note uses first-layer variance $1/(d+1)$, later hidden-layer and readout variances $1/(vN)$, where $v = \mathbb{E}\tanh^2(G)$, and depth $O(d)$. Its uniform Gaussian comparison gives $|p(x \mid S, f) - 1/2| \le \eta$ at every unseen input. Taking $\eta < 1/6$ makes confidence failure exactly the unseen fraction, not merely close to it. Its expected value is again (14). That is a concrete exponential result under the requested confidence criterion, with the specified layer gains and depth; it does not extend to every fixed-depth kernel by changing the prediction rule.

3.4 A small parity eigenvalue does not by itself force slow learning

Proposition 3.9 (A finite-rank Gaussian counterexample)**.** *Fix a nonconstant parity f and $0 < \lambda < 1$. Consider the Gaussian kernel*

$$K_\lambda(x, y) = (1 - \lambda) + \lambda f(x) f(y).$$

It has normalized parity eigenvalue λ. Nevertheless, for its Gaussian sign learner, either with any positive all-input margin or without that margin, expected confidence-failure, posterior, and majority errors after $n \ge 1$ independent uniform labeled examples are at most 2^{-n}. The same bound holds for every zero-temperature limit obtained by shrinking this Gaussian's overall amplitude while keeping the margin fixed.

Proof. Write $G(x) = A + Bf(x)$ with independent $A \sim \mathcal{N}(0, 1 - \lambda)$ and $B \sim \mathcal{N}(0, \lambda)$. There are only two latent values, $A + B$ and $A - B$. With or without the global margin event, the only possible sign functions are $1, -1, f, -f$, each with positive probability. Once the training sample contains both target labels, f is the unique compatible sign function. Otherwise the compatible signs are f and the observed constant, which disagree on half the domain. Every input on the agreeing half has correct-label probability one; hence confidence failure, posterior error, and majority error are all at most $1/2$. The probability that all n target labels coincide is 2^{1-n}. Multiplication yields the claimed bound, uniformly in the Gaussian amplitude, so it persists in any of the specified limits. □

This is a counterexample to an eigenvalue-only inference, not a claim about the iterated-arcsine kernel (6). Its parity eigenvalue can be arbitrarily small while learning remains fast. The counterexample also shows why even an exact, very unfavorable parity prior rate such as (11) does not alone yield a lower bound on examples: a few observations can remove all competitors with larger prior mass.

4 The unresolved exponential parity question

For the precise architecture (1), a fixed kernel does govern every fixed $\gamma < 1$. This follows from exponential freezing, with no substitution of a regression likelihood for the actual sign likelihood.

Three statistical limits result:

$0 < \gamma < 1$	minimum kernel energy over compatible full-margin orthants,
$\gamma = 0$	Gaussian signs conditioned on every margin,
$\gamma < 0$	Gaussian signs conditioned on observed signs alone.

The freezing proof separates the scales N and N^γ. At $\gamma = 1$ they coincide, so the argument cannot discard feature deviations at the scale relevant to learning.

The statements proved here do not establish a uniform $\exp(\Omega(k))$ sample lower bound for k-bit parity throughout $\gamma < 1$. For positive γ the missing step is a quantitative comparison of the minimum energies of compatible sign patterns. For zero or negative γ at fixed depth, it is a quantitative comparison of the relevant Gaussian orthant probabilities; the depth-growing subfamily in Theorem 3.8 is settled here. A regression mean's small correlation with parity, or a small parity kernel eigenvalue, does not supply either comparison. A majority-vote lower bound requires additional control beyond posterior error. These are mathematical gaps in an attempted deduction, rather than a proved counterexample for the specified neural-network kernel.

References

[1] D. Vaintrob and K. Hänni. *A Generalization Result for Overparametrized Neural Net Bayesian Learning*. Lecture notes, 2026. https://kaarelh.github.io/ai/nnbayes-generalization.pdf.

[2] D. Vaintrob, K. Hänni, and ChatGPT. *Small circuits under Gaussian-process classification: what sample complexity follows from the actual posterior?* Unpublished companion note, October 15, 2026.